A210 – Falsifiability: The Four Prediction Quantities

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.1 How the Theory Can Fail

[STIPULATION] A theory that cannot fail at anything says nothing. The A-series therefore carries exactly four quantities at which it can be refuted. They are ratios or structural statements – never pure absolute values, since absolute values are bookkeeping (A080, A130). The first, the τ-mass, is the sharpest: a genuine prediction of a measurable value, because the ratio structure of the leptons binds so tightly.

That there are only four is not a lack of courage, but a consequence of the projection chain (A090): only few quantities survive the projection from the compact structure into the observed world. What disappears in the projection is not measurable – not through ineptitude, but structurally.

.2 The Four Prediction Quantities

[K] First: the τ-mass.

$$m_\tau^{\text{pred}} = 1776.968 \text{ MeV},$$

from (m_e, m_μ) and the structural numbers $(\sqrt{2}, 2/9)$ (A110). Against PDG 1776.86 ± 0.12 that is $+0.90\,\sigma$. A measurement to 0.05 MeV decides: if the value lands near ≈ 1776.97, the pair $(\sqrt{2}, 2/9)$ is confirmed to 10^{-5}; significantly below, it is refuted. This is the sharpest point of the series and its actual prediction core: the theory *does* predict the **lepton masses**. Unlike α or G, here a numerical value stands *before* the measurement – no bookkeeping, no anchor choice, but a genuine prediction, because the ratio structure forces the third mass from two.

[K] Second: the ξ-order of the CHSH deviation. The Bell value falls short of the Tsirelson bound by a residual of order ξ, i.e. 10^{-4} to 10^{-5} (A160). Direction fixed (reduction), order of magnitude fixed (not adjustable), scaling fixed (logarithmic). Not resolvable today; a CHSH value above the bound or significantly below without hardware explanation refutes the theory.

[K] Third: the fractal dimension.

$$D_f = 3 - \xi = 2.99986667.$$

A geometric structure with a measurably different dimension from 3 in exactly this size – or its absence where the theory requires it – decides. D_f is ξ in geometric dress; the deviation from 3 *is* ξ.

[K] Fourth: the fundamental length as anchor point. The definition

$$L_0 = \xi \ell_P$$

places the fundamental length of the theory in relation to the Planck length. Below this sub-Planck scale the theory makes no statement (A130). L_0 is the lower bound of validity of the relative description; a structure that showed ξ-ratios even below it would speak against it.

The numerical value is an anchor matter, not a test. $L_0 = \xi \ell_P$ is the *only* anchor point at which the theory is bound to an absolute scale – and like any anchor, it can be set differently depending on one's viewpoint (against the Planck length, against the Planck time, against $1/\xi$). *Which anchor* is chosen is convention; falsifiable is solely the ξ-scaling: that the boundary lies at ξ times the Planck scale and not at another power.

The numerical value itself does, however, require care – it is not arbitrary. At the standard anchor (Planck length):

$$L_0 = \xi \ell_P = \frac{4}{30000} \cdot 1.616 \cdot 10^{-35}\ \text{m} = 2.155 \cdot 10^{-39}\ \text{m}.$$

An occasionally appearing value $5.39 \cdot 10^{-39}$ m is *not* a valid anchor variant but a transcription error: the mantissa 5.391 is that of the Planck *time* ($t_P = 5.391 \cdot 10^{-44}$ s), and the value implies $\xi = 3.34 \cdot 10^{-4}$ instead of $4/30000 = 1.333 \cdot 10^{-4}$ – a factor 2.5 too large. Convention is the *choice* of anchor; the numerical value at a given anchor is not.

.3 Why They Must Be Ratios

[B] All four are dimensionless or traceable to ξ. This is no coincidence but a principle: only ratios are intrinsic (A080). An absolute value – e.g. "$G = 6.674 \cdot 10^{-11}$" – is bound to a unit choice and can neither confirm nor refute the theory, because it depends on the scale that is itself an input.

That is why α is not on the list, even though A130 treats it in detail: its numerical value is bookkeeping. And that is why the τ-mass is on it, even though it is a mass: what is tested is not its absolute value but the *ratio* that $(\sqrt{2}, 2/9)$ forces between the three lepton masses.

.4 What Is Not a Test Quantity

[STIPULATION] Explicitly *not* counted as falsification:

The agreement of α with $1/137$ (bookkeeping, A130). The numerical value of G (bookkeeping, A145). The ladder ratios at the per cent level (too coarse, check points not predictions, A100). The neutrino sector (not testable, A150). The thesis "SM is a projection" as long as the mapping is missing (A190).

This honesty belongs to falsifiability: listing as a test quantity something one cannot sharply test simulates testability.

.5 Verification Script

- The four prediction quantities with value, comparison quantity, and decision criterion; and the demonstration that each is dimensionless or traceable to ξ: `python/A_Serie_Skripte/a210_prognosegroessen.py`

.6 Open Questions

First, three of the four are not sharply measurable today. Only the τ-mass is within reach of running experiments. The other three are *in principle* testable, but not yet in practice. That is honest to name.

Second, the completeness of the list is not proved. That exactly four quantities survive the projection is an inventory (A090), not a derivation. A fifth is not excluded.

.7 Supersedes

This document collects the test quantities from A090, A110, A120, A130, A160 and presents them as a closed list. It subsumes: Dok. 292 (τ-prediction), Dok. 142 (CHSH), Dok. 101 (L_0, D_f), and the falsification and no-go statements from Dok. 249, Dok. 074 (no-go), and Dok. 193. Incorporated: R52 (numerical value of L_0 at the standard anchor corrected; $5.39 \cdot 10^{-39}$ m rejected as transcription error).

.8 Use of AI Tools

This document was created with the assistance of AI tools. The questions, stipulations, and all substantive decisions come from the author; the tools formulate, compute, and create verification scripts. Responsibility for content and errors rests entirely with the author. Full wording of the rules in A010.